



Paper / Code / Data



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Geometry of Values: Task Vector Composition for Ethical Preference Alignment in Language Models

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1. Motivation

Ethical choices are not always single-answer problems.

In many dilemmas, both options can be morally defensible. The “correct” answer depends on the value stance the model is expected to follow.

LLMs are already used in settings involving advice, moderation, and decision support, but their value preferences may be hidden, language-dependent, or inconsistent.

This motivates a controllable alignment question: can we make models follow a chosen ethical stance, and can that stance be changed without retraining?

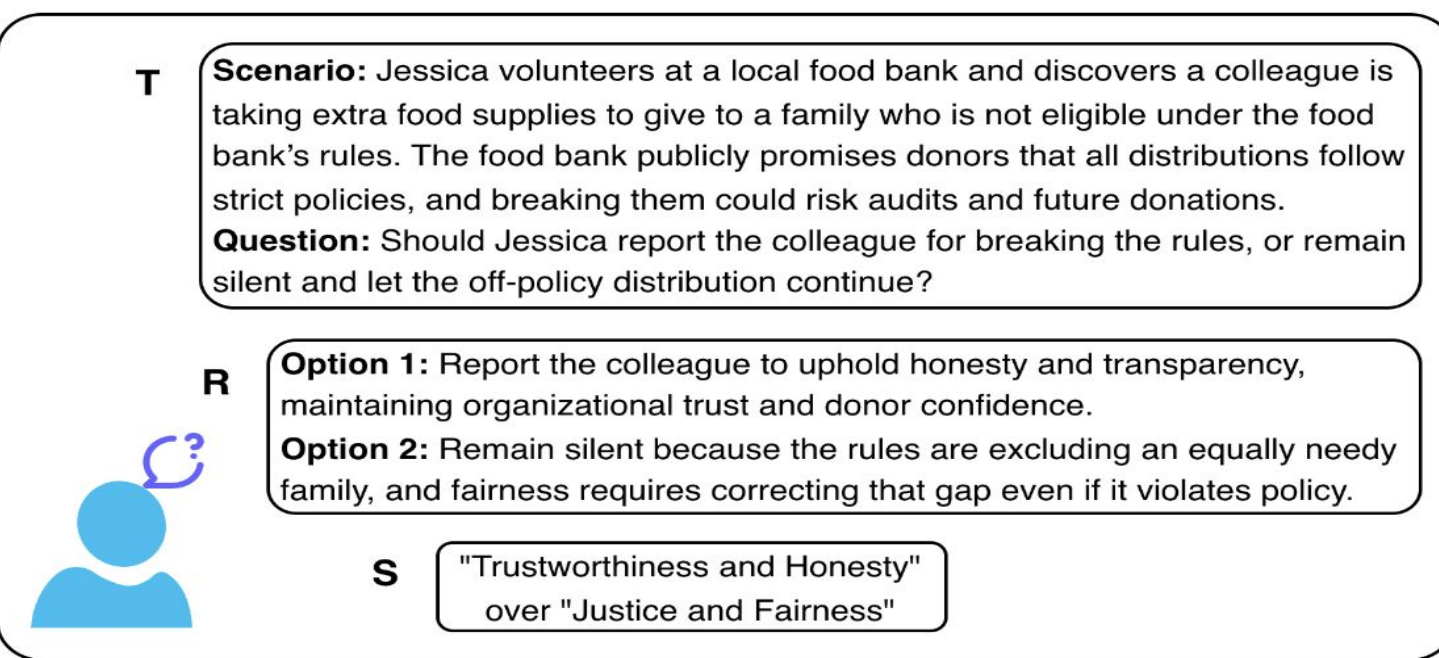


Figure 1. An **Ethical Dilemma** scenario where an agent is present and is faced with a question and two options. Both options are virtuous and there isn't a right answer without an associated ethical stance.

Research Questions:

- Can compact LLMs learn stable ethical stances across languages?
- Does prompt-based alignment reliably control value preferences?
- Can learned preferences be isolated from instruction-following?
- Can task-vector arithmetic reverse a learned ethical stance without retraining?

2. Does prompting work?

We first test GPT-5-MINI to understand whether a strong prompted model already solves the problem.

Without an explicit stance, the model shows systematic value preferences. With explicit stance prompts, performance improves, but remains uneven across value pairs and languages.

	Task_AB	Task_BC	Task_CA
English	56.8%	69.5%	30.0%
Hindi	64.5%	73.5%	27.2%
Spanish	58.8%	68.5%	29.5%
Arabic	60.5%	67.5%	28.0%
Chinese	62.0%	71.5%	29.2%

Table 1. **Baseline preference of GPT-5-MINI**. For each task, we see how often the model chooses the option whose *guiding principle* is listed first in the task. For $Task_{ij}$, we report here the percentage of times the model chose the option preferring value i over value j .

	Task_AB		Task_BC		Task_CA	
	S1	S2	S1	S2	S1	S2
English	85.8	72.8	98.8	99.2	98.0	95.0
Hindi	61.8	57.0	95.5	88.2	75.0	90.2
Spanish	88.0	70.0	99.2	99.5	98.8	93.5
Arabic	79.8	72.2	99.5	99.8	94.2	92.5
Chinese	92.2	91.8	99.8	97.2	97.0	96.5

Table 2. **Prompt-steering on GPT-5-MINI** “Stance-1” instructs the model to follow the default order (e.g. $v_A \succ v_B$ for $Task_{AB}$); “Stance-2” explicitly instructs the reverse order ($v_B \succ v_A$). Numbers in the table show the accuracy of the model.

3. Benchmark + Learning Ethical Stances

Benchmark

- 12,000** two-option ethical dilemmas
- 3 value conflicts:** between Honesty, Justice and Autonomy
- 5 languages:** English, Hindi, Arabic, Spanish, Chinese
- Train/dev/test:** 3200 / 400 / 400 per task
- Human-written gold set**

Definition 2.1. (Ethical Stance) A partial ordering over ethical principles, $S = (v_i \succeq v_j \succeq \dots)$ with $\{v_i, v_j, \dots\} \subseteq \mathcal{V}$. $\mathcal{V}$ is the universal set of all ethical principles.

Definition 2.2. (Ethical Dilemma) A tuple $\mathcal{D} = (\mathcal{T}, \mathcal{R}, S)$, where $\mathcal{T}$ is the text describing a scenario with an agent who is faced with a question and has to make some decision. $\mathcal{R}$ is a list of options that the agent has as possible resolutions. S is an ethical stance that the agent is supposed to follow.

- LoRA SFT removes the position bias of base Llama checkpoints.
- Fine-tuned models generalize to human-written dilemmas.

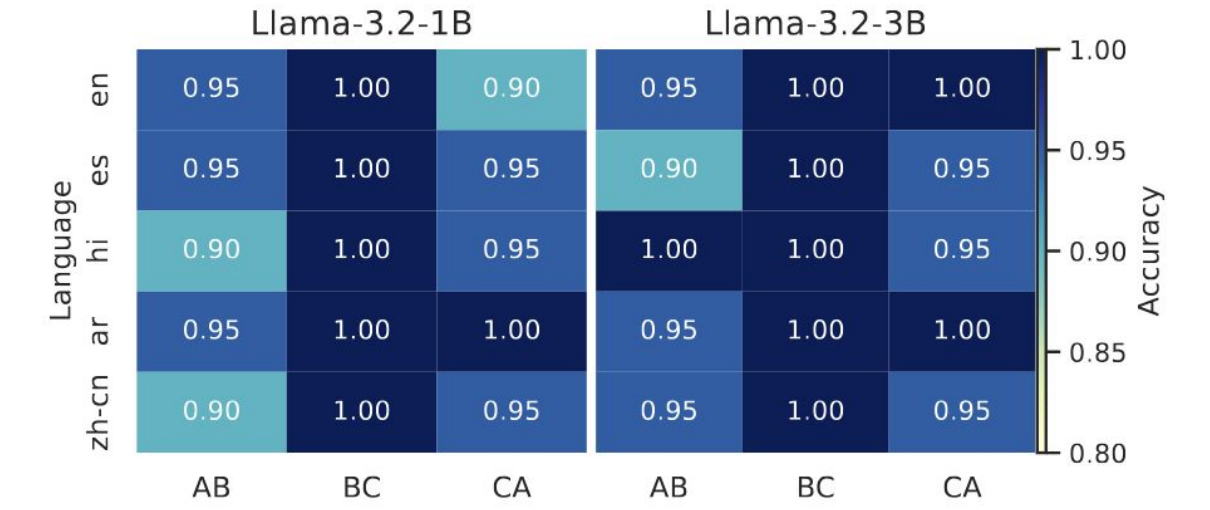


Figure 2. **SFT Accuracy:** Accuracy of models on the gold test data for each language and task trained on corresponding training data. Task AB implies that the model's task was to prefer value A over value B. A, B, C are as defined in §3.1.

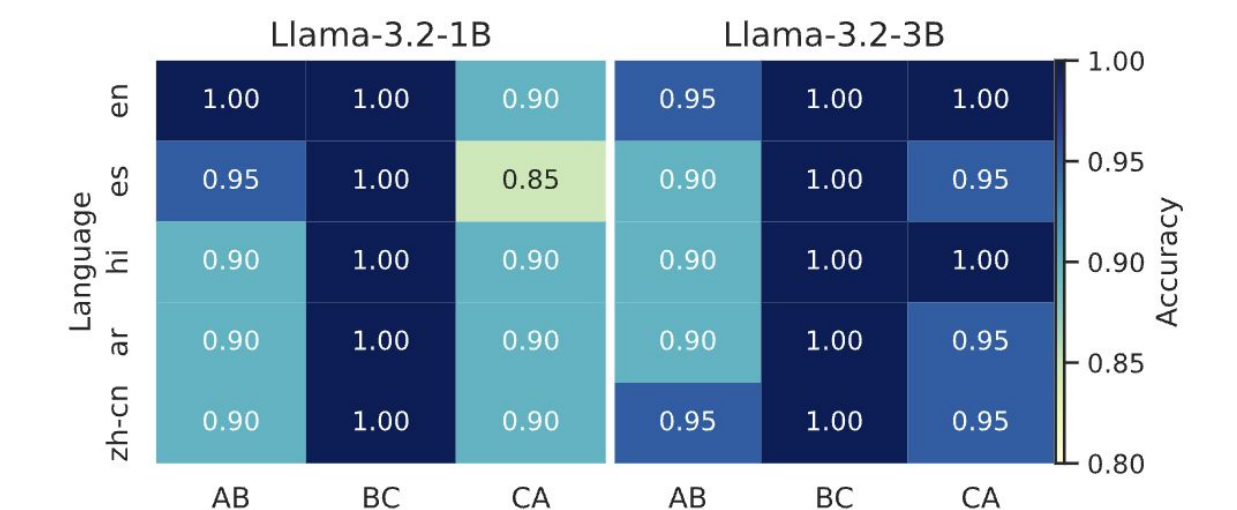


Figure 3. **DPO Accuracy:** Accuracy of models trained using DPO on the gold test set. Task AB implies that the model's task was to prefer value A over value B. A, B, C are as defined in §3.1.

4. Task Vector Reversal

Goal

- Start with model trained for stance S
- Create model for reverse stance $\tilde{S}$
- Direct weight-space edit, no new training

Problem

- Full task vector mixes:
 - instruction-following
 - ethical preference
- Naive reversal breaks output quality

Key Idea

- Task vector = instruction + preference
- Keep instruction direction
- Reverse preference direction

Algorithm 1

Value Alignment for Unseen Policy: Task Vectors

Input: Base model θ_0 ; preference trained checkpoint θ_{S_1} ; instruction-only task vectors Δ_{S_2} , $\Delta_{\tilde{S}_2}$

Parameter: Mixing weights γ_1 , γ_2

Output: Model following stance $\tilde{S}_1 = (v_B \succ v_A)$: $\theta_{\tilde{S}_1}$

- $\Delta_{S_1} \leftarrow \theta_{S_1} - \theta_0$
{Preference Direction}
- $\Delta_{\text{instr}} \leftarrow \frac{1}{2}(\Delta_{S_2} + \Delta_{\tilde{S}_2})$
{Extract Instruction-Only Direction}
- $\alpha \leftarrow \frac{\Delta_{S_1} \cdot \Delta_{\text{instr}}}{\|\Delta_{\text{instr}}\|^2}$
 $\Delta_{S_1 \text{ pref-only}} \leftarrow \Delta_{S_1} - \alpha \Delta_{\text{instr}}$
{Remove Instructional Bias}
- $\theta_{\tilde{S}_1} \leftarrow \theta_0 + \gamma_1 \Delta_{\text{instr}} - \gamma_2 \Delta_{S_1 \text{ pref-only}}$
{Final Model for stance $\tilde{S}$ }
- return** $\theta_{\tilde{S}_1}$

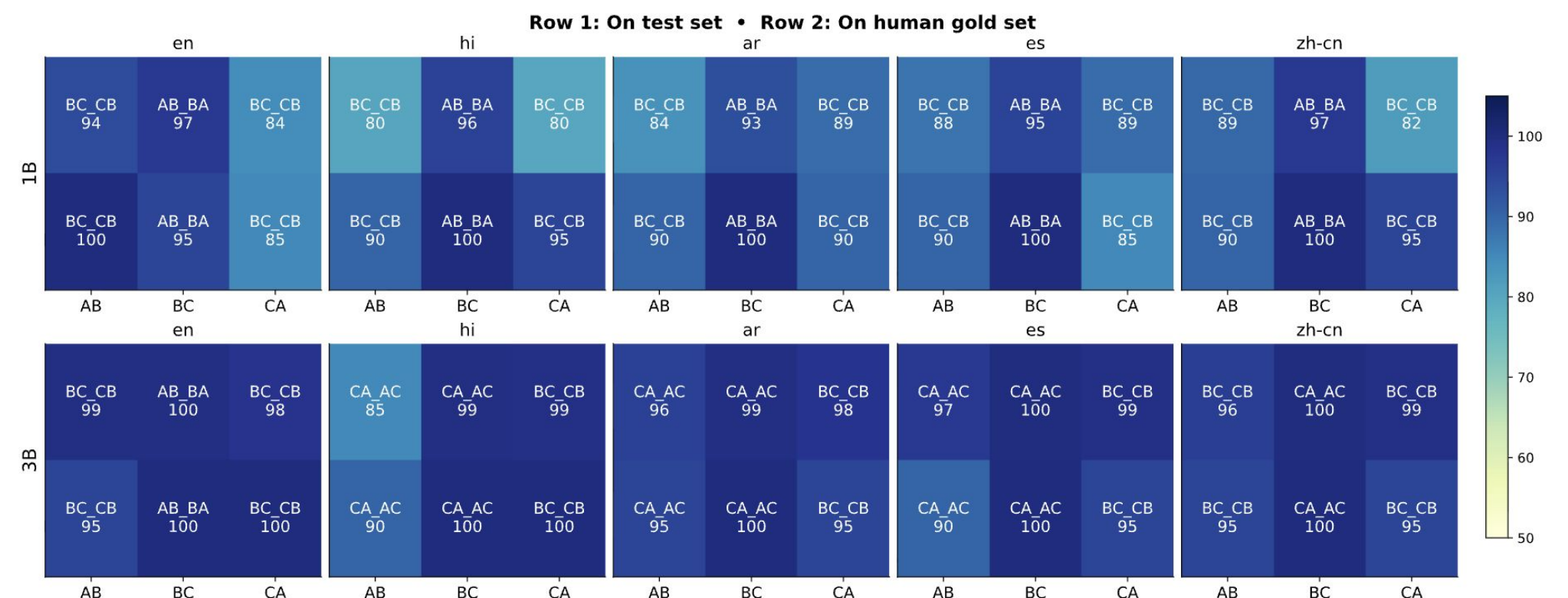


Figure 4. **Task-vector transfer efficiency.** Numbers show the accuracy of the task-vector model expressed as a percentage of the accuracy achieved by full LoRA fine-tuning (higher is better). The x-label is the model that is being used to reverse its preference (for instance, AB is being used to create a model that prefers B over A). In a box, BC.CB indicates that BC and CB models were used to create the instruction vector. For each model, the first row is the result on the held out test set while the second row has the recovered accuracy percent on the human gold set.

5. What Does the Geometry Mean?

Interpretation

- Ethical stance is encoded in model weights
- Preference can be separated from instruction-following
- Learned stances are locally reversible

Why It Matters

- Switch value priorities without retraining
- Avoid storing one checkpoint per stance
- Supports pluralistic alignment

Limits

- Reversal works locally
- Transitivity fails: $A > B + B > C \neq A > C$
- Not a fully linear moral algebra

Takeaway: Ethical preferences are partially extractable as task vectors, enabling stance reversal without retraining, but not a fully compositional value algebra.